

Yao Fan 范尧

CONTACT	<i>Phone:</i> +1 412 452 8917 <i>ORCID:</i> 0000-0002-5660-3201 <i>Personal Website:</i> yao-fan-1	<i>E-mail:</i> yao.fan@pitt.edu <i>PhilPeople:</i> yao-fan-1 <i>PhilPeople:</i> yao-fan-1
AREA OF SPECIALIZATION	Philosophy of AI, Social Epistemology, Theories of Justice, Game Theory, Decision Theory, Philosophy of Economics, Logic	
AREA OF COMPETENCY	Philosophy of Science, History of Modern Philosophy, Political Philosophy, Ethics, Chinese Philosophy	
EDUCATION	University of Pittsburgh , Pittsburgh, PA, United States	
	Ph.D., Philosophy	September 2017 - July 2023
	• Dissertation: Individuals and Institutions: Essays on Social and Political Epistemic Systems (Access through Pitt D-Scholarship) • Committee Members: James Shaw (Co-Chair), Kevin Zollman (Co-Chair, CMU), Kevin Dorst (MIT), Anil Gupta	
	M.Sc., Intelligent Systems (from School of Computing and Information)	
	• Thesis Title: Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning (accepted by IEEE SMC 2023)	
	Australian National University , Canberra, ACT, Australia	
	Ph.D. Candidate, Philosophy (transferred to the U.S.)	February - August 2017
	B.A.(Honours), Philosophy	Feb - Dec 2016
	• <i>First Class Honours</i> • Thesis Title: “Triviality Arguments against Computational Functionalism” • Advisors: Phil Dowe, Alan Hájek, Daniel Stoljar, Koji Tanaka	
	University of Sydney , Sydney, NSW, Australia	
	B.A., Mathematics (Advanced) and Philosophy	March 2014 - December 2015
	• <i>Pass with High Distinction</i>	
EMPLOYMENT	Postdoctoral Researcher , Sun Yat-sen University	September 2025 - August 2027
	Visiting Scholar at Rutgers University Department of Philosophy from June 2026 to May 2027	
	Research Associate , Bridgewater Associates	July 2023 - August 2025
	Research on Global Macroeconomics and AI Applications to Portfolio Management	
	• <i>Bridgewater Associates is the largest hedge fund in the world, participating in global financial market based on fundamental macroeconomic research.</i>	
	Graduate Researcher , University of Pittsburgh	September 2022 - July 2023
	Research on Artificial Social Intelligence	
	• <i>Sponsored by Prof. Michael Lewis (Pitt SCI) and Prof. Katia Sycara (CMU Robotics)</i>	
	Graduate Teaching Assistant , University of Pittsburgh	September 2020 - August 2021

Fully responsible for the teaching and grading for 3 sections, with 20 students in each section.

- *Concepts of Human Nature: this course reads selected writings by Plato, Hobbes, Rousseau, Marx and Freud, with a focus on the historical developments of different conceptions of human nature.*
- *History of Modern Philosophy: this course reads selected writings by Descartes, Locke, Hume and Kant, with a focus on epistemology.*

Graduate Research Assistant, University of Pittsburgh September 2018 - August 2020

Sponsored by Prof. Robert Brandom

- *Research on Expressivist Logic and Dialogical Pragmatics.*
- *Contributed to the editorial work for **A Spirit of Trust** (Harvard University Press, 2019).*

Graduate Teaching Assistant, Australian National University February - June 2017

c Fully responsible for the teaching and grading for 1 section of 30 students.

- *Intermediate Logic: this course covers the semantics and syntax of propositional and first-order logic, with an emphasis on proofs and Logic Programming.*

PUBLICATION

The Inapplicability of Nash Equilibrium to Condorcet Jury Theorem, or I'd rather vote sincerely and so should you

- *Under Review at **Mind***

Regulating Strategic Track-Record Building: Institutional Remedies for Individual Expert Identification (Publisher, Preprint)

- *Synthese, 2026, volume 208, article 11*

Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning (IEEE, arXiv)

- *Li, H., Fan, Y., Zheng, K., Lewis, M., & Sycara, K.*
- *IEEE International Conference on Systems, Man, and Cybernetics, 2023*

SELECTED AWARDS

Global Priority Institute Early Career Fellowship, University of Oxford

- *Award for promising early career researchers in Philosophy and Economics 2021 - 2023*

International Postgraduate Research Scholarship, Australian National University

- *4-year merit-based scholarship awarded to 8 incoming international PhD students each year at ANU 2017-2020 (curtailed after transfer)*

SELECTED TALKS

Systematic Perspectives on Epistemic Deference: Relying on Others and Relying on AI

- *Institute of Philosophy, Chinese Academy of Science, Beijing, China May 2026*

The Epistemic Dimension of Group Decision Making

- *Philosophy Department, Peking University, Beijing, China May 2026*

The Rationality of Sincere Voting

- *MIND Group, Philosophy Department, Rutgers University, United States September 2025*

From Marx to Rawls and Beyond: Or, How I learnt to Stop Worrying and Love “Bourgeois” Theories of Justice

- *Bridgewater Associates Internal Conference, United States* **March 2024**

Centralized and Decentralized Decision-Making

- *Global Priority Institute, University of Oxford, Oxford, UK* **July 2021**

Institutional Remedies for Individual Expert Identification

- *The 4th Early-Career Workshop in Science and Philosophy, Wuhan, China* **June 2021**

Science and Commonsense: An Inferentialist Approach toward Primitivism of Colors

- *12th Arché Graduate Conference, University of St Andrews, U.K.* **February 2021**
- *Shanxi University, Taiyuan, China* **December 2019**
- *Pitt Graduate Work-in-Progress* **October 2019**

From General Logic to Transcendental Logic

- *Pittsburgh Area Kant Colloquium* **October 2020**

Thinking and Being in Plotinus

- *Pitt Graduate Work-in-Progress* **August 2018**

Infinite Gambles in Savage's Decision Theory

- *The 24th World Congress of Philosophy* **August 2018**

Mechanistic Expectations

- *Pitt Graduate Work-in-Progress* **January 2018**
- *Australian Association of Philosophy Annual Meeting, Adelaide, Australia* **July 2017**

Triviality Arguments against Computational Functionalism

- *Philosophy Honours Colloquium, Australian National University* **September 2016**

VISITING

Global Priorities Institute, University of Oxford, Oxford, United Kingdom

Sponsored by GPI Early Career Fellowship **June - July 2021**

Summer School in Logic and Formal Epistemology, Pittsburgh, PA, United States

Hosted by Department of Philosophy, Carnegie Mellon University **June 2018**

Nordic Logic Summer School, Stockholm, Sweden

Sponsored by Nordic Logic Summer School Participation Grant **August 2017**

ANU Logic Summer School, Canberra, Australia

Sponsored by Bob Meyer Memorial Scholarship **December 2016**

Higher School of Economics, Moscow, Russia

Visiting Student specialized in Cognitive Science & Philosophy **September - December 2015**

Peking University, Beijing, China

Graduate Summer School on Epistemology **August 2015**

University of California, Davis, CA, United States

Exchange Student specialized in Mathematics & Philosophy **January - July 2015**

University of California, Los Angeles, CA, United States

Summer Courses on Political Science and Political Philosophy

June - August 2014

PROFESSIONAL
SERVICE

Reviewer

- *Synthese*
- IEEE International Conference on Systems, Man, and Cybernetics (2023, 2026)

University of Pittsburgh

Pitt-CMU Graduate Conference Organizing Committee

- *Member*
- *Co-Chair*

September 2017 - August 2018

September 2018 - August 2019

Pitt Philosophy Mentor Program Organizing Committee

- *Co-Chair*
- *Chair*

September 2017 - August 2018

September 2018 - August 2019

LANGUAGE

Mandarin, English, Classical Chinese, Python, R, Java, Scala, L^AT_EX

OUTREACH

Causation, Causal Direction and Causal Discovery

- *Three Guest Lectures at University of Chinese Academy of Sciences, Beijing, China* **May 2026**

Philosophy Unplugged

- *A series of animated videos about philosophy in Chinese for general audience*
- *Access through Bilibili or YouTube*

Sep 2020 - March 2021

Induction and Causation in Science

- *Two Guest Lectures for Undergraduate Students at Central University of Finance and Economics, Beijing, China* **May 2021**

Science and Society

- *Five Lectures for Advanced High School Students at Veritas China*

July - August 2020

Equality in Contemporary Political Philosophy: Utilitarianism and Rawls

- *Online Public Lecture*

March 2020

GRADUATE
COURSEWORK

Theoretical & Practical Philosophy

Truth and Paradox

Anil Gupta, Fall 2017

Practical Reason

Jed Lewinsohn & Michael Thompson, Fall 2017

† INDICATES AUDIT

Metaphysics and Epistemology

Dmitri Gallow, Spring 2018

Philosophy of Language

Robert Brandom, Fall 2018

Political Philosophy

Japa Pallikkathayil, Fall 2019

Philosophy of Science

General Philosophy of Science

John Norton, Fall 2017

Foundations of Statistics

Teddy Seidenfeld (CMU), Spring 2018

Formal Epistemology of Science †

Kevin Kelly (CMU), Spring 2018

Perception

Mazviita Chirimuuta, Spring 2019

History of Philosophy

Carnap and Quine

Thomas Ricketts, Spring 2018

Philosophy in Late Antiquity: from Philo to Philoponus	Christian Wildberg, Fall 2018
Kant: <i>Critique of Pure Reason</i>	Stephen Engstrom, Fall 2018
Hegel: <i>The Phenomenology of Spirit</i> †	John McDowell, Spring 2019
Kant: <i>Critique of Practical Reason</i> †	Stephen Engstrom, Fall 2019

Economics

Mathematical Methods for Economists 1	Tymofiy Mylovanov, Summer 2019
Mathematical Methods for Economists 2	Sofia Moroni & Roe Teper, Fall 2019
Advanced Microeconomics 1: General Equilibrium	Roe Teper & Luca Rigotti, Fall 2019
Advanced Microeconomics 2: Game Theory	Richard van Weelden, Spring 2020
Evolutionary Game Theory	Kevin Zollman (CMU), Fall 2019
Political Economics †	Richard van Weelden, Fall 2020
Financial Economics	Bryan Routledge (CMU), Spring 2021
Advanced Econometrics 2: Time Series	Yiru Wang, Spring 2023

Logic and AI

Proof Theory	Jeremy Avigad (CMU), Fall 2017
Computability and Definability †	Jeremy Avigad (CMU), Spring 2018
Causal Discovery, Statistics and Machine Learning	Peter Spirtes (CMU), Spring 2019
Algorithm Design	Kirk Pruhs, Fall 2021
Machine Learning	Joseph Yurko, Fall 2021
Natural Language Processing	David Mortensen and Kemal Oflazer (CMU), Fall 2021
Deep Learning	Bhiksha Raj (CMU), Spring 2022
Network Analysis	Morgan Frank, Fall 2022

REFERENCES

Kevin Zollman

Herbert A. Simon Professor of Philosophy
and Social and Decision Sciences
Carnegie Mellon University
kzollman@andrew.cmu.edu

James Shaw

Associate Professor of Philosophy
University of Pittsburgh
James.A.R.Shaw@gmail.com

Kevin Dorst

Associate Professor of Philosophy
Massachusetts Institute of Technology
kmdorst@mit.edu

Robert Batterman (teaching)

Distinguished Professor of Philosophy
University of Pittsburgh
rbatterman@pitt.edu